



Report Title

Public Transport Delay Analysis and Prediction Using Business Intelligence and Machine Learning Techniques & Power BI

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1. Executive summary

This project presents a comprehensive analysis and predictive modeling solution for public transport delays using business intelligence and machine learning techniques. The primary objective of the study was to identify the key factors contributing to delays and develop a predictive system capable of detecting severe disruptions in advance.

The analysis integrates transit delay data from bus, subway, and streetcar systems covering the period 2023 to 2025 with external weather data obtained from the Open-Meteo API. This integration resulted in a unified dataset of over 130,000 observations, enabling a detailed examination of both operational and environmental influences on delay patterns.

Initially, the problem was approached as a regression task to predict delay duration. However, the regression model demonstrated limited explanatory power ($R^2 \approx 0.32$), indicating high variability in delay behavior. Consequently, the problem was reformulated as a classification task, where delays of 15 minutes or more were categorized as severe. This transformation significantly improved predictive performance and aligned the model with operational decision-making needs.

Several machine learning models were evaluated, including Logistic Regression, Random Forest, Balanced Random Forest, and XGBoost. Among these, XGBoost achieved the best overall performance, with an accuracy of approximately 81%, recall of 0.79, and F1-score of 0.68. This indicates a strong balance between identifying severe delays and minimizing false positives.

From a business perspective, the developed solution enables transit authorities to proactively identify high-risk delay situations and take preventive actions such as resource reallocation, schedule adjustments, and operational planning. The findings demonstrate that integrating operational and environmental data significantly enhances predictive capabilities and supports a shift from reactive to proactive transit management.

2. Introduction

This project will focus on analyzing historical weather data to understand patterns, relationships, and trends across key variables such as temperature, precipitation, wind speed, and overall weather conditions. The dataset will consist of time-based observations recorded at regular intervals, allowing for a detailed examination of how weather changes over time. By applying descriptive, diagnostic, predictive, and prescriptive analytics, this project will aim to generate meaningful insights that enhance understanding of weather behavior and support informed decision-making.

Weather variability plays a significant role in both environmental systems and daily activities, as changes in temperature, precipitation, and wind conditions often follow complex and interconnected patterns. These patterns are not always easily identifiable through simple observation alone. This project will demonstrate how a data-driven approach can be used to uncover hidden trends, identify relationships between variables, and provide a deeper understanding of how weather evolves over time.

The analysis will begin by summarizing and exploring the dataset to identify key characteristics and overall trends in the weather variables. It will then investigate relationships between different factors, examining how variables such as wind speed and precipitation may influence temperature and weather conditions. Building on these insights, the project will develop predictive models that will use historical data to estimate future weather patterns, including temperature changes and the likelihood of precipitation. Finally, the project will provide practical insights and recommendations based on the findings, highlighting how historical patterns can be used to anticipate future conditions and improve planning in weather-related contexts.

Overall, this project will illustrate how data analytics can be applied to time-based weather data in a structured and effective way. By combining data exploration, analytical modeling, and visualization techniques, it will demonstrate the value of analytics in understanding complex patterns and generating actionable insights. The methods and findings presented

in this project will also highlight the broader importance of data-driven approaches in environmental analysis and decision-making.

3. Business problem overview

The Toronto Transit Commission (TTC) faces persistent operational challenges related to service delays across its bus, streetcar, and subway networks, which negatively impact passenger satisfaction, service reliability, and overall system efficiency. Delays occur across different routes, stations, and times of day, and are influenced by multiple interconnected factors such as mechanical issues, emergency incidents, peak-hour demand, and operational constraints. These disruptions affect not only travel time but also service consistency across the transit system.

Currently, delay management practices are largely reactive, with operational adjustments often implemented only after disruptions have already occurred. This limits the ability of transit managers to anticipate issues and minimize service interruptions in a timely manner. Given the scale of TTC operations and the frequency of delays across different transport types, there is a growing need for a more proactive and data-driven approach to managing service performance.

A major challenge is the lack of integrated analytical systems that can effectively utilize operational data to identify delay patterns and predict high-risk scenarios. Although large volumes of data are collected regularly, they are not consistently leveraged for advanced analytics, forecasting, or performance optimization. As a result, transit authorities have limited visibility into recurring delay patterns, differences across transport modes, and the root causes of service disruptions.

In addition, the volume and complexity of transit data introduce several data management challenges, including inconsistent entries, missing values, and difficulties in analyzing data across multiple dimensions such as time, location, and transport mode. These challenges

make it difficult to generate timely insights that can support both strategic planning and day-to-day operational decisions.

Addressing these issues requires a structured, data-driven approach that can transform raw operational data into actionable insights. By analyzing historical delay patterns across buses, streetcars, and subways, transit authorities can better understand where and when delays occur, identify key contributing factors, and develop strategies to improve service reliability. This shift from reactive to proactive decision-making has the potential to enhance operational efficiency and provide a more consistent and reliable transit experience for passengers.

4. Analytics questions

To solve the business problem of transit delays in the Toronto Transit Commission (TTC) network, the following analytics questions have been formulated. The questions are organized in a logical sequence, moving from understanding what happened, to why it happened, predicting future delays, and finally recommending actions to reduce delays.

Descriptive Analytics (What happened?)

Question: What percentage of trips are delayed?

Decisions: Helps transit authorities understand the overall frequency of delays and assess the severity of the problem, which can guide whether more detailed analysis or operational improvements are needed.

Question: What is the average arrival delay by transport type (bus, streetcar, subway)?

Decisions: Helps operations managers identify which transport types face the most delays and allocate resources or adjust schedules accordingly.

Question: How do delays vary by weekday and season?

Decisions: Allows planners to understand temporal patterns in delays, supporting schedule adjustments during high-risk periods.

Diagnostic Analytics (Why did it happen?)

Question: How does traffic congestion impact delay duration?

Decisions: Supports operational decisions such as route planning, schedule adjustments, and congestion management strategies.

Question: Do adverse weather conditions increase delays?

Decisions: Helps transit authorities plan preventive measures during extreme weather, such as adjusting schedules or deploying additional vehicles.

Question: Are delays higher during peak hours and public events?

Decisions: Assists operations managers in allocating extra resources or adjusting schedules during peak demand or large events.

Predictive Analytics (What will happen?)

Question: Can we predict whether a trip will be delayed using weather, traffic, and time-based features?

Decisions: Enables planners to anticipate high-risk trips, proactively adjust schedules, or deploy additional resources to prevent delays.

Question: Which variables are the strongest predictors of delays?

Decisions: Helps focus operational and planning efforts on the most influential factors affecting delays, improving efficiency and reliability.

Prescriptive Analytics (What should be done?)

Question: What operational changes can reduce delays during peak congestion or bad weather?

Decisions: Provides actionable recommendations, such as schedule adjustments, reallocation of fleet or staff, and planning services around high-risk periods to improve reliability and customer satisfaction.

5. Scope statement

In Scope:

Analysis of Public Transport Delays:

This project focuses on analyzing delays in the Toronto Transit Commission (TTC) network. We will look at historical trip data to see where and when delays usually happen. The goal is to understand patterns in delays across different routes, stations, and times so we can identify problem areas and see where improvements might be needed.

Impact of External Factors (Temperature, Precipitation, Wind, and Conditions):

We will also study how different weather factors relate to each other.

- **Temperature:** Temperature changes (temperature_2m) will be analyzed to observe variations across time.
- **Precipitation:** Rainfall levels will be examined to identify when and how often precipitation occurs.
- **Wind Speed:** Wind speed (windspeed_10m) will be studied to understand its variation and possible relationships with other weather variables.
- **Weather Conditions:** Weather descriptions (weathercode and weather_desc), such as overcast or clear conditions, will be analyzed to identify frequency and patterns.

Time-Based Effects (Hourly and Daily Patterns):

Time effects, such as hourly variations and daily patterns, will also be considered, since weather conditions change over time and may show recurring trends.

Weather Prediction Using Historical Data:

Another part of the project is using historical data to predict future weather conditions. By studying past trends, we can estimate future temperature, precipitation, and general weather patterns.

Out of Scope:

Real-Time GPS Tracking:

This project will not include real-time tracking of buses or trains using GPS. We will only work with historical data, so live monitoring is not part of this study.

Fare Pricing Optimization:

This project will not look at fare prices or try to optimize them. The focus is only on delays and the factors that cause them.

Infrastructure Construction Planning:

Planning new routes, building stations, or any construction-related activities are not included. The project is limited to analyzing existing operations.

Passenger Behavioral Analysis:

This project will not study how passengers make decisions, their travel habits, or preferences. The focus is strictly on operational data and factors affecting delays.

6. Data sources/key data entities and flows

This project analyzes public transport delays in the Toronto Transit Commission (TTC) system by integrating operational transit data with environmental weather data. The goal is to understand delay patterns across bus, streetcar, and subway services and evaluate how external conditions influence service performance.

A. Data Sources

The analysis uses two primary data sources:

Transit Operations Data (TTC Open Data Portal):

This dataset contains detailed records of delay incidents across bus, streetcar, and subway systems. It is collected in CSV format and includes key attributes such as date, time,

station/location, route/line, delay duration (Min Delay), gap between vehicles (Min Gap), direction (Bound), incident type (Code), vehicle ID, and transport type. This dataset serves as the foundation for analyzing service performance and delay patterns.

Weather Data (Open-Meteo API):

Weather data is obtained from the Open-Meteo API in JSON format and includes hourly observations such as temperature, precipitation, wind speed, and weather conditions. This data is time-stamped and aligned with transit records to examine the influence of environmental factors on delays.

These two sources are integrated to provide a comprehensive view of both operational and external factors affecting transit performance.

B. Key Data Entities

The integrated dataset is structured around three main business entities:

- **Transit Incident:** Represents individual delay events, including attributes such as date, time, station, route, vehicle, and transport type (bus, streetcar, subway).
- **Delay Metrics:** Includes quantitative measures such as Min Delay, Min Gap, direction (Bound), and incident type (Code), which are used to assess delay severity and operational impact.
- **Weather Context:** Captures environmental conditions such as temperature, precipitation, wind speed, and weather descriptions, providing context for analyzing external influences on delays.

These entities collectively support comprehensive analysis and modeling of transit delays.

C. Data Flow Overview

The data flow consists of three main stages:

1. **Data Collection:**

Transit delay data is extracted from the TTC Open Data Portal, while weather data is retrieved from the Open-Meteo API. Both datasets are stored for processing.

2. **Data Integration and Processing:**

The datasets are merged using common time attributes (date and time). During this stage, data cleaning and standardization are performed to handle missing values, ensure consistency, and align formats.

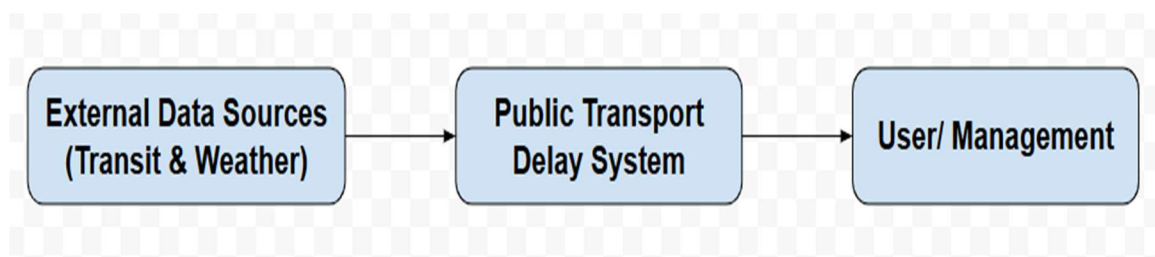
3. **Storage and Analysis:**

The integrated dataset is stored in a structured format and used for analysis, visualization, and predictive modeling to identify delay patterns and trends.

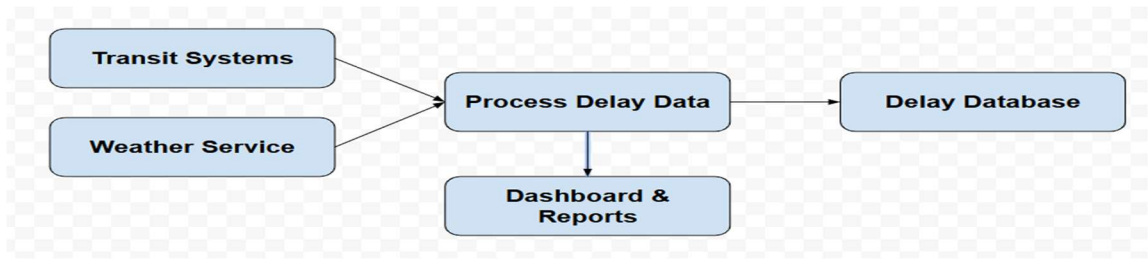
D. Data Flow Diagrams (DFD Summary)

The system can be represented using three levels of data flow diagrams:

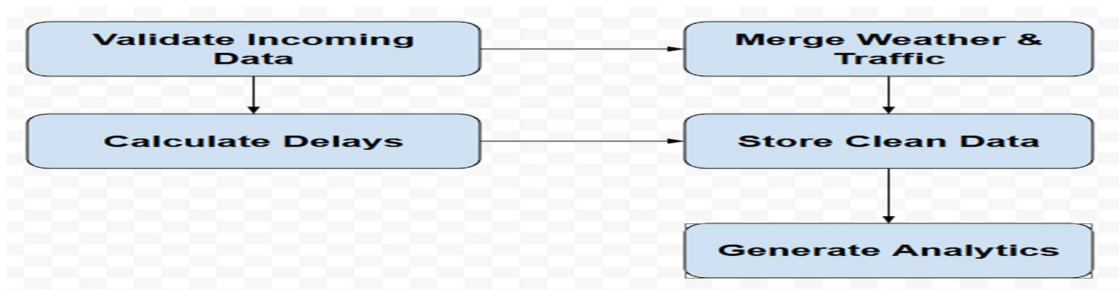
- **Level 0 (Context Diagram):** Level 0 represents the system as a single process where transit and weather data act as inputs, and analytical insights are delivered to users and management.



- **Level 1 (Main DFD):** Level 1 shows how data flows from transit systems and weather services into a processing module, where it is cleaned, stored, and used to generate reports and dashboards.



- **Level 2 (Detailed DFD):** Level 2 illustrates detailed processing steps, including data validation, merging of datasets, delay calculation, storage, and generation of analytics.



These diagrams provide a structured view of how data moves through the system from input to final output.

7. Brief overview of data manipulation process and data output

Data Manipulation Process

The **data manipulation process** in this project was a critical stage in transforming raw and heterogeneous datasets into a structured and analytically meaningful format suitable for machine learning. Since the data originated from **multiple sources**, including transit delay records and external weather data, a systematic approach involving **data consolidation, cleaning, transformation, integration, and feature engineering** was required to ensure consistency, accuracy, and usability.

The process began with **data consolidation across multiple years and transport modes**, where datasets from 2023, 2024, and 2025 covering **bus, subway, and streetcar systems** were integrated. These datasets initially contained **inconsistent formats, varying column structures, and differing naming conventions**, which required standardization. To address this, all datasets were aligned to a **common schema**, ensuring uniformity in variables such as **Date, Time, Line (route), Station, delay metrics (Min Delay, Min Gap), and Transport Type**.

Following filtering and standardization, the data consolidation process resulted in a combined dataset of **184,606 records** across all transport modes. This included **99,040 bus records, 74,395 subway records, and 11,171 streetcar records**. The standardized datasets were then **merged into a unified dataset**, establishing a **single source of truth** that enabled comprehensive system-wide analysis, rather than isolated examination of individual transport modes.

A major challenge encountered during this stage was the **inconsistency in delay categories across datasets**. Earlier datasets used **descriptive labels**, while later datasets relied on **coded values**. To resolve this, a **mapping and standardization process** was implemented to convert all delay codes into **consistent and meaningful categories**. These categories were further consolidated into broader operational groups such as **Mechanical issues** and **Emergency Services**, ensuring **analytical consistency** and reducing noise. This transformation improved both **data interpretability** and **model performance** by eliminating unnecessary complexity.

Following consolidation, the dataset underwent a comprehensive **data cleaning and quality assurance process**. After combining the cleaned subway, bus, and streetcar datasets, the unified dataset consisted of **137,665 records**, forming the initial foundation for analysis. Records with **missing values in critical variables**, particularly **Line (route)** and **Bound (direction)**, were removed to maintain analytical reliability. Additional filtering of **invalid directional values** resulted in a slight reduction to **137,494 records**, indicating **minimal data loss** and effective preprocessing. Additionally, inconsistencies in route identifiers were addressed by applying **regular expression extraction techniques** to isolate numeric route

values. **Invalid or out-of-range routes** were filtered out, ensuring that only **valid operational routes** were retained. Temporal variables were also standardized by combining separate date and time fields into a unified **datetime format**, enabling accurate **time-based analysis** and alignment with external data sources.

To enhance the dataset further, **external weather data integration** was performed using the **Open-Meteo API**. This data included **hourly observations** of temperature, precipitation, wind speed, and weather conditions. Since the weather data was initially obtained in **JSON format**, it was transformed into a **structured tabular format**. A key step involved **mapping numerical weather codes to descriptive categories** based on **WMO standards**, improving interpretability. The weather dataset was then merged with the transit dataset using a **datetime-based join**. The use of a **left join ensured that all original records (137,665)** were retained after integration, preserving **data completeness** while enriching the dataset with environmental variables. This integration enabled the model to capture the **impact of weather on transit delays**, which is a critical factor in real-world operations.

Further enhancement was achieved through **feature engineering**, where new variables were created to capture underlying patterns in the data. Important **temporal features** such as **hour of the day**, **rush-hour indicator**, and **weekday/weekend classification** were introduced to reflect **operational dynamics**. Additionally, **weather-based features** such as **is_rain** and **is_snow** indicators were created to capture **environmental effects**. To address the issue of **high-cardinality route variables**, a **frequency-based encoding approach (Line_freq)** was applied, preserving essential route-level information while reducing dimensional complexity. These engineered features significantly improved the model's ability to detect **nonlinear relationships** and **interaction effects**.

A crucial transformation in the data manipulation process was the **target variable definition**. Initially, the problem was framed as a **regression task** to predict delay duration. However, due to **high variability** and **low predictive performance**, this approach was replaced with a **classification-based formulation**. Delays were categorized as **severe (≥ 15 minutes)** and **non-severe (< 15 minutes)**, creating a **binary target variable**. This transformation was both

technically effective and **business-relevant**, as identifying severe delays is more actionable for decision-making than predicting exact delay values.

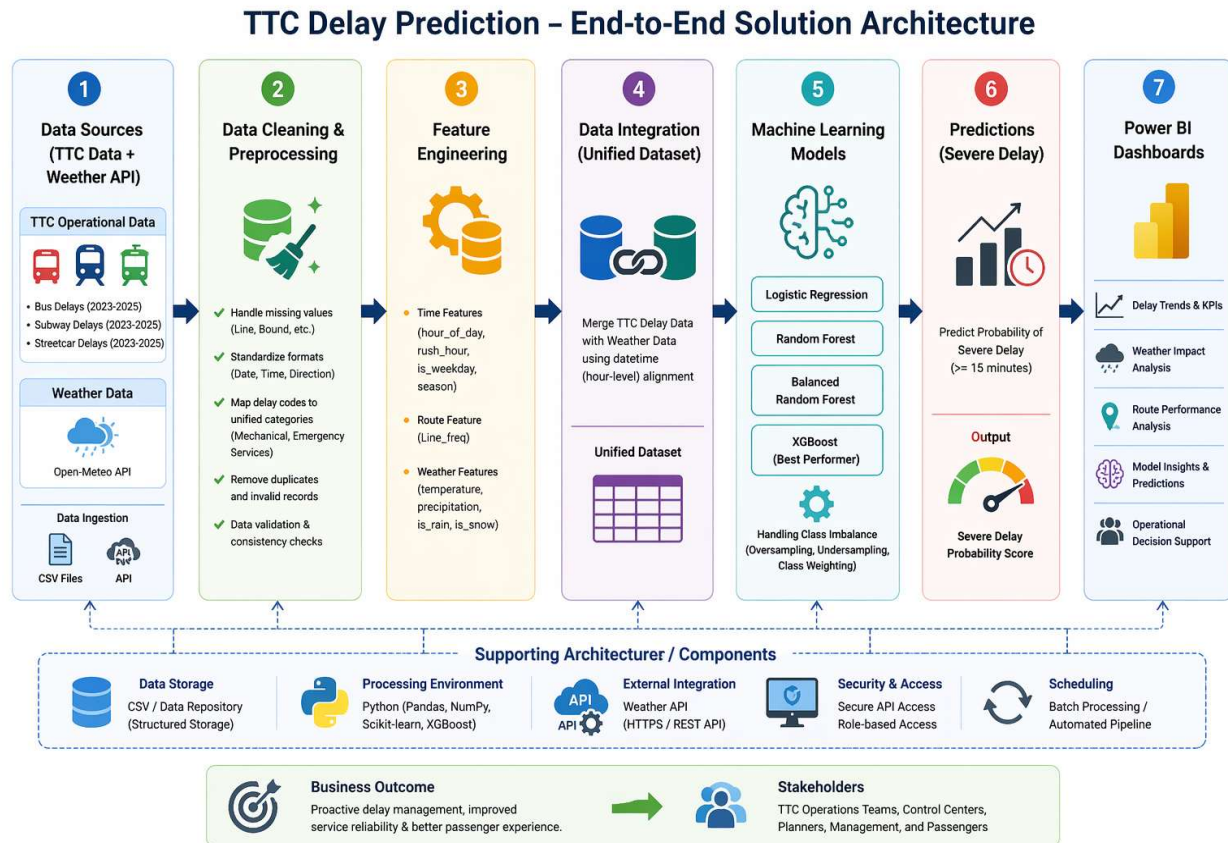
To further improve model robustness, **extreme delay values greater than 60 minutes were treated as outliers and removed**, reducing the dataset from **137,494 to 132,052 records**. This step ensured that the model was not influenced by **anomalous observations** and improved overall **predictive stability**.

The final output of the data manipulation process was a **clean, structured, and fully integrated dataset containing 132,052 records**. This dataset included **standardized transit data, integrated weather variables, engineered features, and a clearly defined target variable**, making it suitable for **machine learning modeling, exploratory analysis, and dashboard visualization**.

Throughout this process, several challenges were addressed, including **data inconsistency, missing values, integration complexity, and high-cardinality variables**. These challenges were resolved through **standardization, filtering, transformation, and encoding techniques**, demonstrating the importance of **robust data preprocessing** in real-world analytics projects.

In conclusion, the data manipulation process not only ensured **data quality and consistency**, but also significantly enhanced the **analytical value** of the dataset. By transforming raw data into a **structured and meaningful format**, this stage enabled the development of **accurate predictive models** and supported **data-driven decision-making** in public transport systems.

8. New solution design and it's fit into the existing IT architecture



The proposed solution is a **data-driven predictive analytics system** designed to enhance TTC delay analysis by integrating key stages such as **data ingestion, data preprocessing, feature engineering, machine learning, and visualization** into a unified pipeline. This solution extends the existing framework by introducing a **predictive modeling layer**, enabling a transition from **descriptive analytics to predictive and prescriptive analytics**. The overall design ensures **data consistency, scalability, and reproducibility**, while maintaining compatibility with the existing IT environment.

The solution begins with the **data ingestion process**, where data is collected from multiple sources including **TTC delay datasets (Bus, Subway, Streetcar: 2023–2025)** and external weather data via the **Open-Meteo API**. This stage combines **batch processing (CSV files)**

with **API-based data acquisition**, allowing integration of both **historical data** and **real-time environmental data**.

Following ingestion, the system performs **data preprocessing and cleaning**, which ensures **data quality and reliability**. This involves removing **missing or inconsistent records**, standardizing **date and time formats**, and cleaning **categorical variables** such as direction and delay codes. Delay codes are mapped into consistent categories such as **Mechanical Issues** and **Emergency Services**, ensuring **data uniformity** across datasets.

The next stage involves **feature engineering**, where new variables are created to enhance **model performance**. This includes **time-based features** such as **hour_of_day**, **rush_hour**, **is_weekday**, and **Season**, along with **route-based features** like **Line_freq**. Additionally, **weather features** such as **temperature**, **precipitation**, **is_rain**, and **is_snow** are incorporated to capture **environmental influences**.

The system then performs **data integration**, where TTC delay data is merged with weather data using **hour-level datetime alignment**. This ensures that each delay record is accurately linked to corresponding weather conditions, resulting in a **centralized and enriched dataset**, which serves as the **single source of truth** for analysis and modeling.

A key enhancement in the proposed solution is the introduction of the **machine learning modeling layer**, where the problem is formulated as a **binary classification task** to predict **severe delays (≥ 15 minutes)**. Multiple models are implemented, including **Logistic Regression**, **Random Forest**, **Balanced Random Forest**, and **XGBoost**. To address **class imbalance**, techniques such as **class weighting**, **oversampling**, and **undersampling** are applied. Model performance is evaluated using **Accuracy**, **Precision**, **Recall**, and **F1-score**, with particular emphasis on **Recall**, as detecting severe delays is critical for effective intervention.

The trained models generate **predictions**, specifically the probability of a **severe delay occurring**, enabling **early identification of high-risk trips**. These predictions support **proactive decision-making** and improve operational responsiveness.

The final stage involves **visualization and reporting**, where the processed data and model predictions are presented through **Power BI dashboards**. These dashboards provide insights into **delay trends, weather impact, model performance, and key performance indicators (KPIs)**, enabling stakeholders to make **data-driven decisions**.

The proposed solution fits seamlessly into the **existing IT architecture** by leveraging current components such as **TTC datasets, Python-based analytics, and Power BI dashboards**. The integration of weather data through **secure API communication (HTTPS)** aligns with existing data acquisition methods, ensuring **compatibility and minimal disruption**.

Furthermore, the introduction of the **machine learning layer** acts as an **extension rather than a replacement**, enhancing analytical capabilities while maintaining the existing technology stack. The reporting layer integrates directly with **Power BI**, allowing **predictive insights** to be embedded into current dashboards and workflows.

Overall, the new solution enhances the existing IT architecture by introducing **advanced predictive capabilities** while maintaining **structural consistency**. It enables the system to evolve from **descriptive and diagnostic analytics to predictive and prescriptive analytics**, supporting **proactive delay management, improved service reliability, and data-driven decision-making**, without requiring major infrastructure changes.

9. New solution implementation and outcome testing

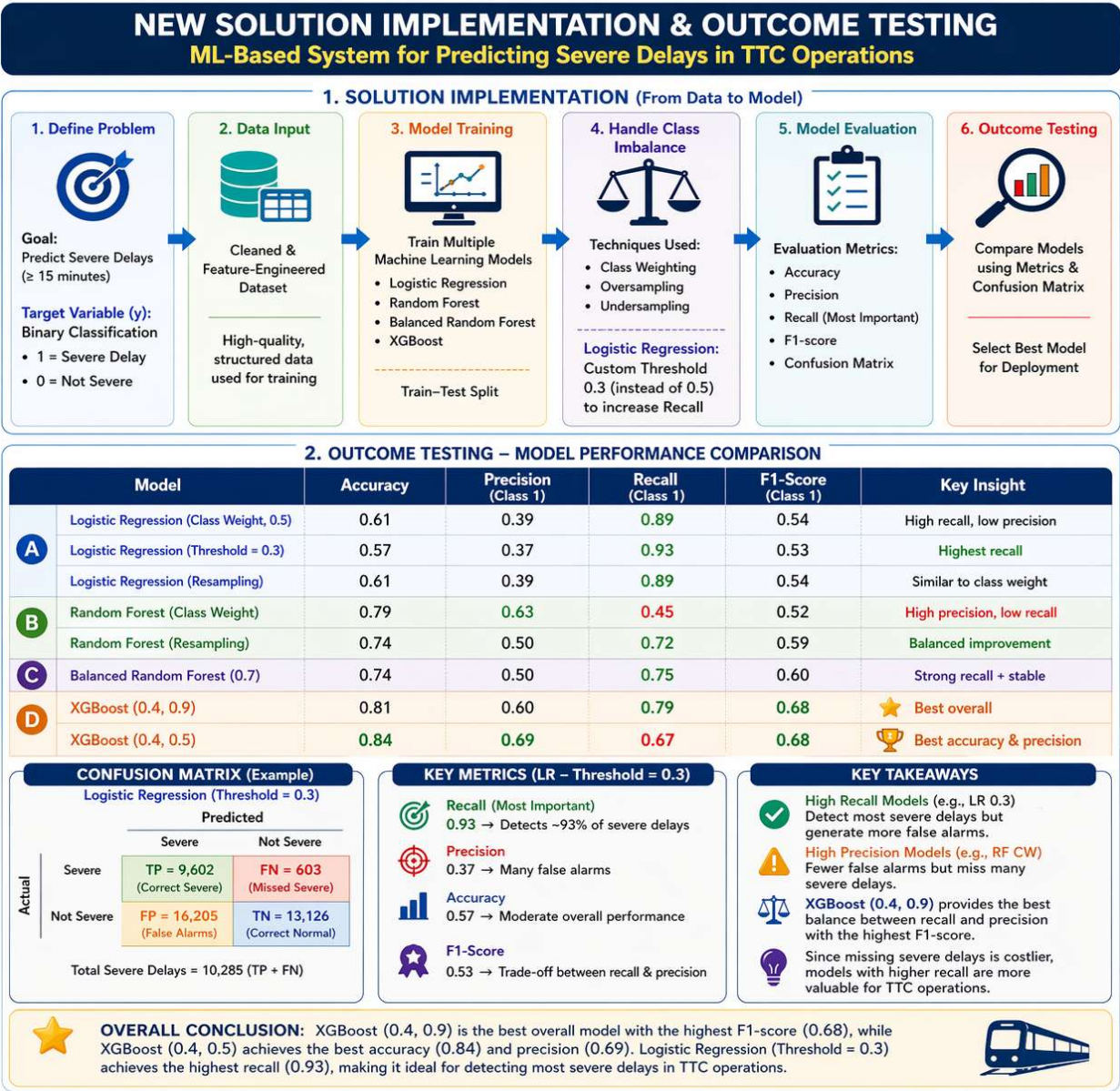
The **new solution implementation** focuses on building and deploying a **machine learning-based predictive system** to identify **severe delays (≥ 15 minutes)** in TTC operations. The implementation begins by defining the **feature set (X)** and the **target variable (y)**, where the target is formulated as a **binary classification variable** indicating whether a delay is severe or not. The **cleaned and feature-engineered dataset** produced in earlier stages is used as input for model training, ensuring that the system operates on **high-quality, structured data**. Multiple **machine learning models**, including **Logistic Regression, Random Forest, Balanced Random Forest, and XGBoost**, are trained and compared to identify the most effective

approach. Since the dataset exhibits **class imbalance**, where severe delays occur less frequently, techniques such as **class weighting, oversampling, and undersampling** are applied to improve the model's ability to detect minority class events. A total of eight models were developed and evaluated using different class imbalance handling techniques. These included **Logistic Regression (Class Weight, Threshold = 0.5)** and **Logistic Regression (Class Weight, Threshold = 0.3)**, where imbalance was addressed using model-based weighting without altering the dataset. A third Logistic model, **Logistic Regression (Scaling + Resampling 0.4, 0.9)**, applied data-level resampling, where **oversampling with a ratio of 0.4 increased the minority class to 40% of the majority class size**, and **undersampling with a ratio of 0.9 reduced the majority class to approximately 1.1 times the minority class size**, creating a more balanced dataset. Similarly, tree-based models included **Random Forest (Class Weight)** and **Random Forest (Resampling 0.4, 0.9)** using the same resampling strategy. In addition, a **Balanced Random Forest (Sampling Strategy = 0.7)** was implemented, where the algorithm internally resampled the minority class to **70% of the majority class size** during training. Finally, two boosting models, **XGBoost (Resampling 0.4, 0.9)** and **XGBoost (Resampling 0.4, 0.5)**, were developed, where the latter used stronger undersampling (**0.5**, meaning the majority class was reduced to twice the minority class size), enabling comparison of moderate versus aggressive balancing strategies. These models allowed for a comprehensive evaluation of class weighting, resampling, and threshold tuning approaches in improving the prediction of severe transit delays.

Following model training, the system proceeds to the **outcome testing phase**, where performance is evaluated using key **evaluation metrics**, including **Accuracy, Precision, Recall, and F1-score**, along with the **Confusion Matrix**, which provides a detailed breakdown of **True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN)**.

For instance, in the case of **Logistic Regression with a threshold of 0.3**, the confusion matrix shows that the model correctly identifies **9,602 severe delays (TP)** while missing only **683 severe delays (FN)**, resulting in a very high **Recall of approximately 0.93**. This indicates that the model successfully detects **93% of all severe delays**, which is critical for operational

effectiveness. However, the model also produces **16,205 false positives (FP)**, leading to a relatively low **Precision of approximately 0.37**, meaning that many predicted severe delays are not actually severe. The overall **Accuracy of around 0.57** reflects moderate performance due to the high number of false alarms.



positives indicates that the system may generate excessive alerts, which could affect operational efficiency if not properly managed. Comparative outcome testing across models reveals distinct performance differences. The **Random Forest model with class weighting** achieves higher **precision** but significantly lower **recall (~0.45)**, meaning it misses many severe delays and is less suitable for this use case. The **Balanced Random Forest model** improves recall to approximately **0.75**, offering more stable and balanced performance by internally addressing class imbalance. In contrast, **XGBoost emerges as the best overall model**, achieving a strong balance between **precision and recall**, along with the highest **F1-score**, making it the most suitable model for deployment in a real-world setting.

Overall, the implementation and outcome testing demonstrate that the proposed solution can effectively predict severe delays using **machine learning techniques**, while also providing flexibility to adjust model behavior through **threshold tuning and model selection**. The integration of **evaluation metrics, confusion matrix analysis, and model comparison** ensures that the final solution is both **technically robust and practically applicable**, enabling **data-driven and proactive decision-making** in TTC operations.



10. Potential solution optimization

The proposed solution demonstrates strong capability in predicting severe delays in TTC operations; however, there are several opportunities for further **optimization** to enhance its **accuracy, robustness, scalability, and real-world applicability**. These improvements focus on key areas such as **feature engineering, model optimization, class imbalance handling, data enrichment, evaluation strategies, and system deployment**, ensuring that the solution evolves into a more advanced and reliable predictive system.

One of the most impactful areas for improvement is **feature engineering**, as the quality and relevance of features directly influence model performance. While the current solution includes **time-based features** such as *rush_hour* and *is_weekday*, along with **weather-based features** like *temperature* and *precipitation*, additional features can be introduced to capture more complex temporal and operational patterns. For instance, incorporating **lag-based features**, such as previous delays on the same route or at the same time, can help the model learn sequential dependencies. Similarly, **rolling statistics** (e.g., average delays over the past few hours or days) can provide contextual information about delay trends. The creation of **interaction features**, such as *weather × rush hour* or *route × time of day*, can further help capture **non-linear relationships** that are not easily detected by basic features. These enhancements would allow the model to better represent real-world conditions and improve predictive accuracy.

Another critical area of optimization is **model tuning and selection**. Although **XGBoost** has demonstrated the best overall performance in the current implementation, its performance can be further improved through **hyperparameter tuning**. Techniques such as **Grid Search, Random Search, or Bayesian Optimization** can be used to optimize parameters like *learning rate, max depth, number of estimators, and subsampling ratios*. Proper tuning can significantly improve both **model generalization** and predictive performance. In addition, exploring alternative algorithms such as **LightGBM and CatBoost** may provide computational efficiency and improved handling of categorical variables, especially for

larger datasets. Ensemble approaches, such as **stacking or blending multiple models**, could also be considered to combine the strengths of different algorithms and further improve performance.

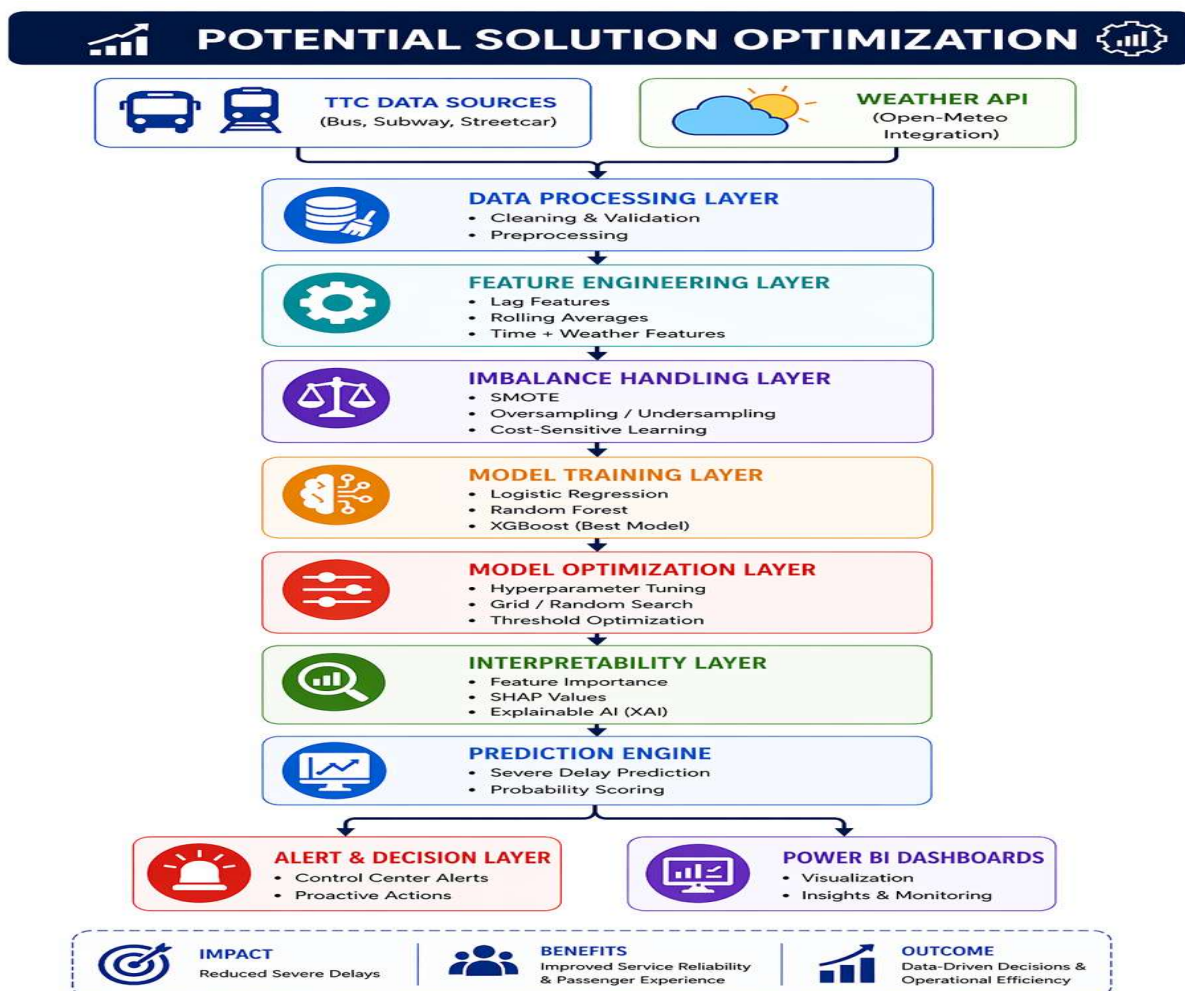
Handling **class imbalance** more effectively is another important optimization opportunity. In the current solution, techniques such as **class weighting, oversampling, and undersampling** are used to address imbalance between severe and non-severe delays. However, more advanced methods such as **SMOTE (Synthetic Minority Oversampling Technique)** can be implemented to generate synthetic samples of the minority class, helping the model learn better decision boundaries. Additionally, **cost-sensitive learning** can be applied to assign higher penalties to misclassifying severe delays, ensuring that the model prioritizes critical cases. Combining resampling techniques with ensemble methods, such as **Balanced Bagging or boosting-based approaches**, can further enhance the model's ability to detect rare but important events.

The **evaluation and threshold optimization process** can also be refined to improve decision-making. Currently, a fixed threshold (e.g., 0.3 in Logistic Regression) is used to improve **recall**, but this approach may not always provide the optimal balance between recall and precision. A more systematic approach using **ROC curves, Precision-Recall curves, and cost-based optimization** can help determine the most suitable threshold based on business requirements. For instance, if the cost of missing a severe delay is significantly higher than generating a false alarm, the threshold can be adjusted dynamically to prioritize recall. This ensures that model performance aligns closely with operational objectives.

Another important optimization area is **data enrichment and quality improvement**. While the current solution integrates TTC delay data with weather data, additional external data sources can further enhance prediction accuracy. For example, incorporating **traffic conditions, road incidents, special events, public holidays, and infrastructure disruptions** can provide a more comprehensive view of the factors influencing delays. Improving the **granularity of weather data**, such as using location-specific or higher-frequency data, can also lead to more precise predictions. Furthermore, implementing continuous **data**

validation and cleaning pipelines ensures that the model is always trained on high-quality and up-to-date data.

From a system perspective, optimizing the **deployment and scalability** of the solution is essential for real-world implementation. Transitioning from a batch-processing system to a **real-time or near real-time prediction system** can significantly improve operational responsiveness. This can be achieved by integrating the model into a **streaming pipeline or API-based system**, allowing predictions to be generated continuously as new data arrives. Additionally, implementing **automated model retraining pipelines** ensures that the model adapts to changing patterns over time, maintaining accuracy and relevance. Integration with **alert systems and dashboards** can further enhance usability by providing timely notifications to control centers.



Finally, improving **model interpretability and explainability** is crucial for building trust and enabling effective decision-making. Techniques such as **feature importance analysis, SHAP (SHapley Additive exPlanations) values, and model explainability tools** can be used to identify the key factors influencing predictions. This helps stakeholders understand why certain delays are predicted, making the system more transparent and actionable. Improved interpretability also supports better operational planning by highlighting the most significant drivers of delays.

11. Assessment of Descriptive, Diagnostic, Predictive, and Prescriptive Analytics

The analytics questions formulated for this project were designed to address the business problem of transit delays in the Toronto Transit Commission (TTC) network through a structured approach encompassing **descriptive, diagnostic, predictive, and prescriptive analytics**. The overall solution integrates both **Power BI dashboards for data visualization** and **machine learning models for prediction**, resulting in a comprehensive analytics framework. The project successfully addressed these questions, with particularly strong performance in **descriptive and predictive analytics**, while diagnostic analysis was partially limited by the availability of external data sources.

The **descriptive analytics questions**, which focus on understanding *what happened*, were comprehensively addressed through both **exploratory data analysis (EDA)** and the **Power BI dashboard**. The dashboard provides clear insights into the **percentage of delayed trips**, including metrics such as **Severe Delays (%)** and **On-Time vs Delayed distribution**, enabling a direct understanding of delay frequency and severity. Additionally, the analysis of **average arrival delay by transport type (bus, streetcar, subway)** was effectively visualized using bar charts, allowing easy comparison across transport modes. Temporal variations in delays were also captured through multiple dashboard visuals, including **monthly trends, weekday vs weekend comparisons, seasonal analysis, and rush hour vs non-rush hour patterns**. These visualizations, supported by engineered features such as **is_weekday, hour_of_day**,

and rush_hour, provided a strong and intuitive understanding of delay patterns across time and operations, establishing a solid descriptive foundation.

The **diagnostic analytics questions**, which aim to explain *why delays occur*, were also addressed using both **feature engineering and dashboard analysis**, particularly in relation to **weather and time-based factors**. The integration of external weather data from the **Open-Meteo API** enabled the evaluation of how conditions such as **rain, snow, and clear weather** impact delays. Dashboard visualizations comparing **rain vs no rain, snowfall vs no snowfall, and clear vs non-clear weather** revealed that weather conditions did not have a statistically significant impact on delay duration. Similarly, the analysis of **rush hour vs non-rush hour delays** showed minimal variation, suggesting that delays are more influenced by **operational factors** rather than environmental conditions. However, certain diagnostic aspects were only partially addressed. Specifically, **traffic congestion** was not directly measured due to the absence of traffic data, and the impact of **public events** was not explicitly included in the analysis. Despite these limitations, the combination of weather, temporal features, and dashboard insights provided meaningful understanding of key delay drivers.

The **predictive analytics questions**, which focus on *what will happen*, represent one of the strongest components of the project. The objective of predicting whether a trip will experience a **severe delay (≥ 15 minutes)** was successfully achieved using multiple **machine learning models**, including **Logistic Regression, Random Forest, Balanced Random Forest, and XGBoost**. These models were trained on a dataset enriched with **time-based, route-based, and weather-based features**, enabling accurate classification of severe delays. Model performance was rigorously evaluated using **accuracy, precision, recall, F1-score, and confusion matrix analysis**, with particular emphasis on **recall**, as detecting severe delays is critical for operational decision-making. Techniques such as **class imbalance handling (oversampling, undersampling, and class weighting)** and **threshold tuning (e.g., 0.3 for Logistic Regression)** were applied to improve model effectiveness. Furthermore, the

inclusion of **model interpretability techniques**, such as feature importance analysis, helped identify the most influential predictors, strengthening the analytical insights.

The **prescriptive analytics questions**, which focus on *what should be done*, were effectively addressed by combining insights from both the **dashboard and predictive models**. The results suggest that **high-recall models**, such as Logistic Regression with threshold tuning, are suitable for detecting the majority of severe delays, while **XGBoost provides the best overall balance between precision and recall**, making it ideal for deployment. The system design supports **real-time prediction and alert generation**, enabling transit control centers to take **proactive actions**, such as adjusting schedules, reallocating vehicles or staff, and preparing for high-risk conditions. Additionally, the dashboard enables continuous monitoring and supports **data-driven decision-making**, enhancing operational efficiency and service reliability.

12. Conclusion

The project successfully addressed the majority of the analytics questions across all four analytical stages by integrating **data visualization and machine learning techniques**. Descriptive and predictive analytics were comprehensively implemented, diagnostic analysis provided meaningful insights despite some data limitations, and prescriptive analytics delivered actionable recommendations. While future improvements could include incorporating **traffic data and public event information** to strengthen diagnostic capabilities, the current solution demonstrates a **robust, practical, and end-to-end approach** to understanding, predicting, and managing transit delays in the TTC network.